

SHASHWAT MAHAJAN

+91-93569 97907 | shashwatmahajan2@gmail.com | linkedin.com/in/shashwatmahajan2 | github.com/Shashwat-Mahajan

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), C, SQL

AI / ML: Large Language Models (LLMs), LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Embeddings, ChromaDB, HuggingFace, Groq API, XGBoost, Scikit-Learn, Pandas, Streamlit

Web & Backend: React.js, Next.js, Node.js, Express.js, FastAPI, MongoDB, PostgreSQL, WebSockets, REST APIs

DevOps & Tools: Docker, GitHub Actions, CI/CD Pipelines, Git, Linux, JWT

EXPERIENCE

GenAI Engineer Intern

Jan 2026 – Present

Generative AI Division

- Designing and deploying production LLM-powered features using LangChain, RAG pipelines, and prompt engineering workflows — work ships directly to end users in a live environment.
- Optimising vector retrieval pipelines for latency and response accuracy across multiple Generative AI systems.
- Owning end-to-end GenAI feature delivery: prompt design, vector store tuning, deployment, and post-launch monitoring.

EDUCATION

Yeshwantrao Chavan College of Engineering, Nagpur

Expected June 2027

B.Tech in Computer Science Engineering | CGPA: 8.00 / 10

Relevant Coursework: Machine Learning | Data Structures & Algorithms | Database Systems

PROJECTS

DevOnboard — AI-Powered Repository Intelligence Platform

IBM Hackathon

Python | LangChain | RAG | Vector Embeddings | GitHub API | LLM

- Built an AI platform that ingests any GitHub repository via URL, parses its structure, README, and source code, and returns cited, context-aware answers to developer onboarding questions — eliminating hours of manual codebase exploration.
- Engineered a chunking-and-embedding retrieval pipeline using semantic vector search to deliver accurate Q&A across large, multi-file codebases with full source attribution.
- Showcased at IBM Hackathon; solution compresses new-developer ramp-up time by automating codebase comprehension end-to-end.

Financial Analyst RAG System

Python | LangChain | ChromaDB | Groq API | HuggingFace Embeddings | Streamlit | AWS S3 | GitHub Actions | RAGAS

- Built a production-grade Retrieval-Augmented Generation (RAG) pipeline ingesting live financial news from multiple URLs, returning cited and contextually accurate query responses for real-time multi-company analysis; persists ChromaDB vector store to AWS S3 for durable, cost-efficient retrieval across sessions.
- Automated continuous ingestion of latest financial data via GitHub Actions scheduled workflows, ensuring the knowledge base stays current without manual intervention; enforced role-based access control (RBAC) to restrict query access by user tier.
- Deployed as a live Streamlit application with secure API key handling, retrieval caching, and graceful error fallback; implemented HuggingFace query-expansion for dynamic cross-company comparisons, LLM guardrails for safe output, and RAGAS evaluation framework for ongoing retrieval quality benchmarking.

Credit Risk Assessment System

React.js | FastAPI | XGBoost | Docker | GitHub Actions | CI/CD

- Architected a full-stack ML inference system for real-time credit risk predictions: React.js frontend → FastAPI backend → XGBoost classifier, serving low-latency results suitable for production traffic.
- Trained XGBoost model on 10,000+ records achieving 88% accuracy; applied SMOTE oversampling to correct a severe class imbalance and eliminate majority-class prediction bias.
- Containerised with Docker and implemented a GitHub Actions CI/CD pipeline, enabling zero-downtime deployments with automated test coverage on every commit.

Smart Attendance System — Smart India Hackathon (SIH)

Team Lead | 6-Member Team

MongoDB | Express.js | React.js | Node.js | Supabase | PostgreSQL | WebSockets

- Led a 6-member cross-functional team for SIH National Hackathon: owned system architecture, sprint planning, task delegation, and live demo delivery to judges.
- Designed rolling QR-based authentication to prevent replay attacks; enforced MFA and Supabase Row-Level Security (RLS) for strict per-user data isolation.
- Architected a real-time WebSocket synchronisation layer supporting 100+ concurrent users with zero data loss under sustained load.

HACKATHONS

SITNOVATE — Real-Time Chat Engine

24-Hour Hackathon

React | WebSocket | JWT

- Designed and shipped a JWT-secured real-time chat system entirely within 24 hours; scoped architecture under severe time constraints and delivered a live working demo to a judging panel.

ACHIEVEMENTS & CERTIFICATIONS

- LeetCode Rating: 1538 — 400+ problems solved | CodeChef: 2-Star Coder | Primary Languages: Python & Java
- Machine Learning Specialization — Codebasics.io | Full Stack Web Development (MERN Stack)